

SOFIA BOTTI

Curriculum Vitae

Personal Information

Date of birth: September 1995
Nationality: Italy
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Current Position

01/2026 - ... **Contract Researcher**
Politecnico di Milano, Department Mathematics, MOX
realized via FIS grant SYNERGIZE, Grant agreement ID: FIS-2023-02228
Milano (Italy)

Employment history

08/2025 - 01/2026 **Lecturer of Module "Algorithmics"**
UniDistance Suisse, Faculty of Mathematics and Computer Science
Brig (Switzerland)

11/2023 - 10/2025 **Postdoctoral Researcher**
Università della Svizzera Italiana
supported by SNSF grant, Grant agreement ID: 217025
Lugano (Switzerland)

11/2022 - 10/2023 **Postdoc Assistant**
University of Pavia and Università della Svizzera Italiana
realized via H2020-EuroHPC Microcard project, Grant agreement ID: 955495
Pavia (Italy) and Lugano (Switzerland)

Education

11/2022 **PhD in Computational Mathematics and Decision Sciences**
Joint PhD Program UNIPV-USI, 2019 – 2022
Thesis: *Mathematical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs): from ionic currents to 3D ventricle models.*
Supervisor: Prof. Luca F. Pavarino

07/2019 **Master in Mathematics**
University of Pavia, Department of Mathematics, Pavia (Italy) 2017 – 2019
Thesis: *Mathematical and numerical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs).*
Supervisor: Prof. Luca F. Pavarino
Mark: 110/110 *cum laude*

07/2017	Bachelor in Mathematics University of Pavia, Department of Mathematics, Pavia (Italy) 2014 – 2017 Thesis: <i>Sistemi dinamici: equivalenza topologica ed esempi di biforcazioni - Dynamic Systems: Topological Equivalence and Examples of Bifurcations.</i> Supervisor: Prof. Giuseppe Savaré Mark: 103/110
06/2014	Secondary school certificate Scientific Lyceum Giovanni Gandini, Lodi (Italy) 2009 – 2014 Mark: 100/100 <i>cum laude</i>

Fundings and grants

11/2023-10/2025	Swiss National Science Foundation (SNSF), Postdoctoral Fellowship grant , <i>Cell-by-cell and bidomain models for cardiac derived stem cell tissue: innovative numerical and deep learning tools for regenerative medicine</i> , PI: Sofia Botti, Grant number: 217025, approx. 211 450 CHF
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Scientific visiting periods

11/2025	King Abdullah University of Science and Technology (KAUST) , Thuwal (Saudi Arabia) Host: Prof. Rolf Krause
11/2024	Tampere University , Tampere (Finland) Host: Prof. Jari Hyttinen Project: Chamber-Specific Ionic Models in Regenerative Cardiology: Bridging Single-Cell Precision to Tissue-Level Heterogeneity
04/2022-06/2022	Ente Ospedaliero Cantonale (EOC) - Cardiocentro Ticino (CCT) , Bellinzona - Lugano (Switzerland) Host: Prof. Lucio Barile and Dr. Claudia Altomare Project: Dynamic Clamp setup and action-potential recording in stem cells
04/2021-10/2021	Università della Svizzera Italiana (USI) , Lugano (Switzerland) Host: Prof. Rolf Krause Project: Optimization tools for the calibration of the atrial-specific ionic model

Supervision of junior researchers

Co-supervised Ph.D. students:

09/2024-...	Tonali Sofia , <i>Computational modeling of hiPSC-CMs MEAs</i> , Ph.D. student in <i>Computational Mathematics, Learning, and Data Science</i> at Università degli Studi di Pavia
04/2025-...	Granda Alexandra , <i>CardioTwin: Precision Cardiology based on Digital Twins</i> , Ph.D. Student at Università della Svizzera Italiana

Co-supervised Master theses:

10/2025-...	Rorri Foteini , <i>Learning Inverse mappings in electrophysiology at the micron scale via the EMI model</i> , MSc in Artificial Intelligence, Università della Svizzera Italiana
10/2023-4/2024	Tonali Sofia , <i>Mathematical modeling of Multielectrode Arrays with Pluripotent Stem Cell-Derived Cardiomyocytes</i> , MSc in Mathematics, Università degli Studi di Pavia

Teaching activities

Lecture list

08/2025-01/2026	Algorithmics , Department of Mathematics, (BSc, fall semester), FernUni-Unidistance, Brig, Switzerland
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Lecture list (Lecturer in Ph.D. courses)

06/2025	Modeling the electrophysiology of the cardiac tissue , Intensive Summer School on Computational Cardiology: theory, applications and new trends, Milano (Italy)
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Lecture list (Substitute lecturer)

10/2022-01/2023	Sistemi dinamici - Substitute Lecturer , Department of Industrial engineering and information and Department of Mathematics (MSc), Università degli Studi di Pavia, Pavia, Italy
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Lecture list (Teaching assistant)

10/2022-01/2023	Analisi 1 , Department of Industrial engineering and information (BSc), Università degli Studi di Pavia, Pavia, Italy
10/2021-01/2022	Analisi 1 , Department of Industrial engineering and information (BSc) and Department of Civil engineering and Architecture (BSc), Università degli Studi di Pavia, Pavia, Italy
12/2020-02/2021	Analisi 1 - Didactic seminars , Department of Industrial engineering and information (BSc), Università degli Studi di Pavia, Pavia, Italy
11/2020-12/2020	Mathematics - Precourse , Faculty of nursing, Università degli Studi di Pavia, Pavia, Italy

Prizes, awards and fellowship

09/2024	Travel Grant for the Cardiac Physiome Workshop 2024 – 500€
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Organization of conferences and events

09/2025	Organizer, Mini-symposium <i>Recent Advances in Mathematical and Computational Methods for Cardiac Biomechanics</i> , at Enumath 2025, Heidelberg (Germany)
06/2025	Co-organizer, <i>Intensive Summer School on Computational Cardiology: theory, applications and new trends</i> , at Università degli Studi di Milano (Italy)
05/2024	Co-organizer, ICAM Workshop, <i>International Center for Advanced Computing in Medicine</i> , at Università della Svizzera Italiana (Switzerland)

Personal skills

Languages	Italian (mother tongue), English (Fluent), French (Basic)
Operating systems	Linux, macOS, Microsoft Windows
Programming languages	MATLAB, C/C++, Python
Technical tools	bash, Vim, Git, Docker
Visualization tools	Paraview
Documentation	L ^A T _E X/TikZ/Pgfplot/Beamer, Microsoft office, Keynote

PUBLICATIONS

SOFIA BOTTI

Publications

Journal Articles:

- [J5] **S. Botti**, M. Favino, R. Krause, C. Altomare, L. Barile, A Correlation Analysis of Field Potential and Action Potential Biomarkers in Heterogeneous hiPSC-derived Cardiac Tissues. *In preparation*,
- [J4] **S. Botti**, M. Favino, The Cost of Accuracy: Evaluating Bidomain and MEA Models for Cardiac Electrophysiology. *Submitted paper*
- [J3] **S. Botti**, R. Krause, L. F. Pavarino (2025). *In-silico* Modeling of Multi-Electrode Arrays for Enhancing Cardiac Drug Testing on hiPSC-CMs Heterogeneous Tissues. *Journal of Physiology*, Published article
- [J2] **S. Botti**, C. Bartolucci, C. Altomare, M. Paci, L. Barile, R. Krause, L. F. Pavarino, S. Severi (2024). A novel ionic model for matured and paced atrial-like hiPSC-CMs integrating IKur and IKCa currents. *Computers in Biology and Medicine*, Preprint, Published article
- [J1] **S. Botti**, M. Torre (2023). Isogeometric simulation of a derived stem cell engineered ventricle. *Advances in Computational Science and Engineering*, 1(3), 298-319. Published article

Peer-reviewed Conference Proceedings, Book Chapters:

- [C4] S. Tonali, C. Altomare, L. Barile, R. Krause, L. F. Pavarino **S. Botti**. Efficient Hybrid IMEX Schemes for Modeling Multi-Electrode Arrays in Stem Cell-Derived Cardiac Tissue. *Submitted proceeding*
- [C3] **S. Botti**, R. Krause, L. F. Pavarino (2026). Advanced Bidomain Framework for Drug Testing in Heterogeneous Cardiac hiPSC Tissues. *Accepted paper in Cardiovascular Modeling, in Springer INdAM series*
- [C2] **S. Botti**, C. Bartolucci, R. Krause, et al. (2023). An in silico Study of Cardiac hiPSC Electronic Maturation by Dynamic Clamp. In *Functional Imaging and Modeling of the Heart*, 171-183. Published article
- [C1] **S. Botti**, C. Bartolucci, et al. (2022). Numerical Simulations Indicate IK1 Dynamic Clamp Can Unveil the Phenotype of Cardiomyocytes Derived from Induced Pluripotent Stem Cells. In *2022 Computing in Cardiology (CinC)*, Published article

PhD thesis:

- [T1] **S. Botti** (2022) Mathematical modeling of Human Induced Pluripotent Stem Cell-Derived Cardiomyocytes (hiPSC-CMs): from ionic currents to 3D ventricle models, PhD thesis, Published thesis

Presentations

Plenary and invited talks:

- [T14] **S. Botti**, M. Favino, Multiscale Computational Modeling of Heterogeneous hiPSC-CM Cardiac Tissues. **Invited speaker** at International Workshop on Numerical and Learning Methods for PDEs, Sanya (China), 2026.
- [T13] **S. Botti**, M. Favino, Multiscale Computational Modeling of Heterogeneous hiPSC-CM Cardiac Tissues. **Invited seminar** at KAUST (Saudi Arabia), 2025.
- [T12] **S. Botti**, Chamber-Specific Ionic Models in Regenerative Cardiology: Bridging Single-Cell Precision to Tissue-Level Heterogeneity. **Invited seminar** at MET International Seminar Series, Tampere (Finland), 2024.
- [T11] **S. Botti**, *In-silico* modeling of multi-electrode arrays to enhance cardiac drug testing on heterogeneous hiPSC-CMs tissues. **Selected oral presentation** at Cardiac Physiome, Freiburg (Germany), 2024.
- [T10] **S. Botti**, Mathematical and Numerical Modeling of cardiac stem cells and regenerative medicine, **Invited speaker** at CompMath2023 Spring Workshop, Pavia (Italy), 2023.

Conference talks:

- [T9] **S. Botti**, M. Favino, Steklov-Poincaré domain decomposition for hiPSC-CMs multi-electrode arrays models. 29th International Conference on Domain Decomposition Methods (DD29), Milano (Italy), 2025.
- [T8] **S. Botti**, R. Krause, L.F. Pavarino, Modeling hiPSC-CM Electrophysiology on Multi-Electrode Arrays: A Coupled Bidomain and Electrode Approach. Coupled Problems, Villasimius (Italy), 2025
- [T7] **S. Botti**, A novel atrial-specific ionic model for stem cells-derived CMs. 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (WCCM-PANACM), Vancouver (Canada), 2024.
- [T6] **S. Botti**, A novel atrial-specific ionic model for stem cells-derived CMs. (INdAM Workshop), Rome (Italy), 2024.
- [T5] **S. Botti**, Numerical modeling of cardiac derived stem cells and isogeometric simulation of an engineered tissue. European Conference on Numerical Mathematics and Advanced Applications (ENUMATH), Lisbon (Portugal), 2023.
- [T4] **S. Botti**, M. Torre, Isogeometric solvers for cardiac reaction-diffusion equations coupled with Induced Pluripotent Stem Cell ionic models, 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo (Japan), 2023.
- [T3] **S. Botti**, C. Bartolucci, C. Altomare, et al., Numerical Simulations Indicate IK1 Dynamic Clamp Can Unveil the Phenotype of Cardiomyocytes Derived from Induced Pluripotent Stem Cells, Computing in Cardiology (CinC), Tampere (Finland), 2022.
- [T2] **S. Botti**, From cardiac stem cell ionic models to isogeometric simulations of 3D cardiac tissue, CompMath2022 Spring Workshop, Pavia (Italy), 2022.
- [T1] **S. Botti**, M. Torre, Isogeometric simulations of hiPSC-CM cardiac tissue, Young Researchers Workshop on Mathematical and Numerical Cardiac Modeling, Pavia (Italy), 2021.

Posters:

- [P3] **S. Botti**, M. Favino, et al., A Correlation Analysis of Field Potential and Action Potential Biomarkers in Heterogeneous hiPSC-derived Cardiac Tissues, LS2 Cardiovascular Research Meeting 2026, Bern (Switzerland), 2026.

- [P2] **S. Botti**, R. Krause, et al., Modelling Atrial Fibrillation in Stem Cell-Derived CMs: *in-silico* approaches for Personalized Medicine, Gordon Research Conference (GRC), Lucca (Italy), 2025.
- [P1] **S. Botti**, C. Bartolucci, R. Krause, et al., An *in silico* Study of Cardiac hiPSC Electronic Maturation by Dynamic Clamp, Functional Imaging and Modeling of the Heart (FIMH), Lyon (France), 2023.

Other products with documented use

- [G1] **MEA model**, Open-source MATLAB framework for 2D Multi-Electrode Array (MEA) simulations of heterogeneous hiPSC-CM cardiac tissues. [Code repository](#) (Co-core developer)
[DOI:10.5281/zenodo.18593287](https://doi.org/10.5281/zenodo.18593287)
- [G1] **AL_hiPSC_ionic_model**, Open-source MATLAB code for the atrial-specific ionic model for human induced pluripotent stem cells. [Code repository](#) (Core developer)
[DOI:10.5281/zenodo.18589161](https://doi.org/10.5281/zenodo.18589161)